

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)

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Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

Brown dwarf atmospheres contain patchy, inhomogeneous clouds. As a brown dwarf rotates, these clouds periodically rotate into and out of view, blocking or revealing the flux beneath them and producing periodic brightness variations; local temperature changes can further affect the observed flux. Studying this variability helps reveal the cloud structure of brown dwarf atmospheres and how it evolves.

We selected the target 2MASSW J0036159+182110 because it is the brightest object in I-band among our candidate sample, ensuring a high signal-to-noise ratio and a precise lightcurve. Its rotation period is known to be about 3 hours. We plan to observe it continuously for about 8 hours using the 90-inch telescope at Kitt Peak, with 200-second exposures in the I band. Given the ~3-hour rotation period, an 8-hour run should, if successful, cover roughly two complete rotation cycles.

With this observation, we hope to obtain a high-precision I-band lightcurve, determine the amplitude and period of the variability, and check whether the lightcurve remains consistent across the two cycles

Telescope Requested (0.5 points): 90"

Instrument (0.5 points): 90 Prime

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

We think the object we want to observe should keep 50 deg away from the moon.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension (deg)	Declination (deg)	V magnitude (i-band and r-band)	Object Type	Visibility (on Sep 30th)

2MASSW J0036159+1 82110	9.0671	18.3529	17.1922 19.5066	L3.5	~ 8.67h
2MASSW J0149090+2 95613	27.2875	29.9368	17.7745 19.9889	M9.5	~8h

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

~8h, 200s exposure time. $SNR \sim (\text{flux} * 200s)^{(1/2)}$, we can find flux by our mag value. In the paper Aren N. Heinze_2014, they use 200s for all BD. So 200s can also give us a suitable SNR in this project.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

The 8-hour run covers about two full rotation cycles of the ~3-hour period, allowing us to confirm the periodicity and check whether the variability amplitude and shape stay consistent from one cycle to the next. Continuous 200-second exposures give us dense time sampling, so the lightcurve shape and period can be measured accurately rather than just roughly estimated. Choosing the brightest target in I-band ensures high signal-to-noise, so the measured variability amplitude is reliable rather than lost in noise. Together, these choices let us directly measure the target's rotation period, variability amplitude, and any cycle-to-cycle changes — the main goals stated in the abstract.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (0.5 points)

If clouds are present but not total, we can still proceed with the observation, since relative photometry (measuring the target flux relative to nearby reference stars) is largely insensitive to uniform cloud extinction — previous similar surveys have obtained useful photometry through clouds causing up to 50% extinction. In this case, our backup plan is to continue monitoring through thin or intermittent clouds rather than cancelling the run, accepting a somewhat noisier lightcurve.